Mark schemes

Q1.

H_0: No association between choice of subject and gender

H_1: Association between choice of subject and gender

B1

	Bul	**Cl**	**Fin**	**Pol**	**Total**
Male	7	31	25	40	103
Female	2	24	22	19	67
Total	9	55	47	59	170

Totals

B1

O_i	E_i
7	5.45
2	3.55
31	33.32
24	21.68
25	28.48
22	18.52
40	35.75
19	23.25
170	**170**

E's attempted (correctly)

M1A1

One of the E_i's < 5 $\therefore$ combine cells

Attempt at combining (correctly)

M1A1

O_i	E_i	$\alpha = (O_i - E_i)$	α^2/E_i

47	41.20	5.8	0.8165
21	26.80	−5.8	1.2552
31	33.32	−2.32	0.1615
24	21.68	2.32	0.2483
25	28.48	−3.48	0.4252
22	18.52	3.48	0.6539

Final column

m1

Test statistic: $X^2 = 3.56$

(AWFW 3.55 to 3.57)

A1

Critical value: = 4.605

ft on their ν

B1F

Accept H_0

A1F

Insufficient evidence to suggest that the choice of subject is associated with gender.

E1

11

[11]

Q2.

H_0 : no association between age and attitude to school reorganisation

H_1 : association between age and attitude to school reorganisation

B1

Age	Against	
	O_i	E_i
16 – 17	9	6 $^{17}/_{65}$
18 – 21	17	15 $^{24}/_{65}$
22 – 49	115	116 $^{9}/_{13}$
50 – 65	41	42 $^{9}/_{13}$
> 65	3	3 $^{64}/_{65}$
Total	185	185

Age	Not Against

	O_i	E_i
16 – 17	2	$4\ {}^{48}/_{65}$
18 – 21	10	$11\ {}^{41}/_{65}$
22 – 49	90	$88\ {}^{4}/_{13}$
50 – 65	34	$32\ {}^{4}/_{13}$
> 65	4	$3\ {}^{1}/_{65}$
Total	140	140

E's attempted correctly (at least 6 E's)

E_i
6.262
15.369
116.692
42.692
3.985

E_i
4.738
11.631
88.308
32.308
3.015

M1A1

Row totals: **11,27** 205, **75.7** (325)
Column totals: 185, 140 (325)

Totals correct

B1

E_i's < 5
∴ combine cells 16 – 17 and 18 – 21 **also**

Attempt at combining rows

M1

50 – 65 and 'over 65' to give:

Correctly

A1

O_i	E_i	$\alpha = O_i - E_i$	α^2/E_i
26	21.63	4.369	0.8825
115	116.69	−1.692	0.0245
44	46.68	–2.677	0.1535
12	16.37	–4.369	1.1662
90	88.31	1.692	0.0324
38	35.32	2.677	0.2029
325	325		2.462

Final column attempted (dep M1)

m1

$X^2 = 2.462$

2.4 to 2.5

A1

$v = 2$

B1

$\chi^2_{v=2}(0.95) = 5.991$

On their v

B1ft

Accept H_0

A1ft

No real evidence at 5% level of significance to suggest any association between age and attitude to school reorganisation.

(context)

E1ft 12

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Q3.

(a) H_0: Choice independent of gender

gender not associated with choice

B1

	Squash	Badminton	Archery	Hockey	
Male	5/3.5	16/14	30/24.5	19/28	
Female	4/5.5	20/22	33/38.5	53/44	M1

Combine Squash and Badminton

E_i < 5 (Similar categories)

M1

	S & B	Archery	Hockey	
Male	21/17.5	30/24.5	19/28	M1
Female	24/27.5	33/38.5	53/44	M1

χ^2 values

	S & B	Archery	Hockey
Male	0.7000	1.2347	2.8928
Female	0.4455	0.7857	1.8409

M1

χ^2calc = 7.90

(7.8 to 7.9)

A1

$v = 2$

B1

$\chi^2_{5\%}(2) = 5.991$

(on their v)

B1ft

10

Reject H_o

Sufficient evidence, at the 5% level of significance, to support an association between the choice of sport and gender

reject H_o and H_o stated

or

statement in context

(b) **More females and fewer males** chose

B1

to participate in hockey **than expected**

B1

2

[12]

Q4.

(a)

	D		P		DN	
< 65	4	2.70	12	7.72	6	11.58
≥ 65	10	11.30	28	32.28	54	48.42

Method for E's

M1

Pooling D&P

	D&P		DN	
< 65	16	10.42	6	11.58
≥ 65	38	43.58	54	48.42

correct method for pooling - must be D&P

m1

H_0 : Answer not associated with age

B1

H_1 : Answer associated with age

correct null hypothesis - may be implied by clearly stated conclusion

$$\frac{(|O-E|-0.5)^2}{E} =$$

attempt at $\sum \frac{(O-E)^2}{E}$ *- their figures*

M1

attempt at Yates' correction

m1

correct method for Yates'

m1

$$\frac{5.08^2}{\left(\frac{1}{10.42}+\frac{1}{11.58}+\frac{1}{43.58}+\frac{1}{48.42}\right)} = 5.83$$

5.83 (5.81 to 5.85)

A1

c.v χ_1^2 is 3.841

1df or 2df if no pooling

B1

3.841 or 3.84 (allow 5.991/5.99 if no pooling)

B1

Reject H_0, conclude evidence to suggest that answer is associated with age.

correct conclusion - must be compared with upper tail of χ^2

A1ft

10

(b) Older delegates more likely to answer that they would not change their votes.

Older delegates less likely to answer that they would change their votes

E1

1

(c) Most delegates are over 65

Most delegates are elderly

E1

1

[12]

Q5.

(a)

	0	1	2	≥3	Total
<2	1478 1508.94	212 191.24	$\underbrace{21 \quad 6}_{27}$ $\underbrace{14.36 \quad 2.45}_{16.81}$		1717
>2	2830 2799.06	334 374.76	$\underbrace{21 \quad 1}_{21}$ $\underbrace{26.64 \quad 4.55}_{31.19}$		3185
Total	4308	546	41	7	4902

method for 4 E's

M1

Method for all E's

M1

Attempt to combine classes

M1

Correct method for combining classes - needs all previous M marks

m1

H_0 : No association between number of claims and time for which policy held.
H_1 : Number of claims associated with time for which policy held.

Correct H_0 - may be implied by correctly stated conclusion.

B1

$$\sum \frac{(O-E)^2}{E} = \frac{30.94^2}{1508.94} + \frac{30.94^2}{2799.06} +$$

Attempt at $\sum \frac{(O-E)^2}{E}$ *- allow spurious continuity correction etc*

M1

$\frac{20.76^2}{191.24} + \frac{20.76^2}{354.76} + \frac{10.19^2}{16.81} + \frac{10.19^2}{31.19}$

Correct use of $\sum \frac{(O-E)^2}{E}$ *- requires first two and 4th M mark*

m1

= 14.0

14.0 (13.9 to 14.1)

A1

c.v. χ_2^2 for 1% risk is 9.21

2 df

B1f.t.

9.21, disallow ± 9.21

B1f.t.

Reject H_0 . Conclude number of claims associated with time policy held.

Conclusion - must be compared with upper tail of χ^2*.*

A1f.t.

11

(b) (i), (ii) **A**

Observed members not frequencies

E1

	Claim	No claim
Male	412	2824
Female	181	1485

cao

M1

A1

B

Classes overlap

E1

	Claim	No claim
<25	280	676
≥ 25	314	3632

cao

M1

A1

C

Classes not complete

E1

9

	No claim	≤ £2000	>£2000
<25	676	135	145
≥ 25	3632	122	192

cao

M1

A1

[20]

Examiner reports

Q1.

This topic, as usual, proved to be the best source of marks on the paper, especially for those candidates who realised that when the *expected* frequency of any one category falls below 5 then this category *must* be combined with another. When combining, this should have been done with the category to which it was most closely related. In this case, combining Bulgarian with Polish to form a new category ‘Languages’ was the most sensible and, thankfully, the most often seen. Unfortunately, too many candidates simply combined the adjacent categories of Bulgarian and Climate Change or failed to combine at all.

A few candidates combined for the wrong reason by stating that “*observed* frequencies less than 5”, or combined categories without having shown any evidence that expected frequencies had been calculated. The null hypothesis was usually stated correctly and the correct conclusion in context was often seen. However, some candidates were unable to state the correct critical value found from the χ^2 tables, whilst a few thought incorrectly that the use of a t-value or a z-value was appropriate. Some candidates, when quoting H_0, seemed to think that the expressions ‘not associated’ and ‘not independent’ had the same meaning.

Q2.

This topic, as usual, proved to be the best source of marks on the paper, especially for those candidates who realised that when the expected frequency of any one category falls below 5 then this category must be combined with another. When combining, this should have been done with the category to which it was most closely related. In this case, the 16-17 and 18-21 age groups should have been combined, and also the 50-65 and Over 65 age groups, since each of these pairs of categories probably contained people with fairly similar and quite closely related views on the reorganisation of schools. It was not thought that combining the 16-17 age group with the Over 65 age group was sensible in the context of this question. Some candidates only combined once, either at the top end or at the bottom end of the age range and it was quite disappointing to see others not combining any categories at all with the consequential loss of marks.

The null hypothesis was usually stated correctly and the correct conclusion in context was often seen. However, some candidates were unable to state the correct critical value found from the χ^2 tables, whilst a few thought incorrectly that the use of a t-value or a z-value was appropriate.

Q3.

This question produced many disappointing responses. In part (a), most candidates stated correct hypotheses and usually worked out the required values for the E_i. However, there were still some candidates who insisted on rounding these values to the nearest whole number; which is penalised. The calculations of the E_i. were usually followed by an incorrect assumption that $\nu = 3$ and consequently the wrong χ^2 value was quoted from the tables. Although many candidates failed to realise that when E_i. < 5, similar categories must be combined, the majority went on to use the correct method for evaluating χ^2. The conclusions drawn were usually correct and often in context. However, some candidates still think, incorrectly, that the rejection of the null hypothesis proves that the alternative hypothesis must be true. Part (b) proved to be beyond all but the very best candidates. The vast majority failed to interpret their results found in part (a). Most candidates simply referred to the original data, usually stating that 'more females than males played hockey' rather than making the required reference to the expected and observed values.

Q4.

As usual candidates were well able to deal with contingency tables although some lost marks by failing to pool. A few pooled without first calculating any expected values, implying that they had based their decision to pool on the observed rather than the expected values.

Q5.

Candidates performed very well on this question. Part (b) was testing, but most candidates were able to make a reasonable attempt.